Pre-Feasibility Study for Sumac Cultivation And Production Project

2017

Table of Contents

List of Tables

List of Figures

1. Executive summary

The project aims to cultivate and produce sumac in Tafilah governorate by selecting an area of 20 donum to cultivate the desired varieties of local sumac , in addition to preparing, grinding ,milling and packaging Sumac in plastic bags with a capacity of (100-1000 gr) or carton , also can be sold by weight but unpacked. Sumac is a shrub with fallen pendulous branches that reaches a height of about two meters, wide extended branches, and its leaves are in pairs. Its flowers are in groups of red and blueberry fruits in a redish color. Ruby Red. The used parts of sumac plant are; ripe fruits peel, seeds and dried leaves, as for the commercial production of sumac needs 3-4 years from the beginning of agriculture, the project lifetime was assumed to be 15 years. Sumac is used fresh and sprinkled over the food to add flovers to regular Salad and particularly to (Fattouch),. Moreover, Sumac is considered as a basic material to prepare thyme and the popular dish made of chicken and bread (Al-Msakhan).

Table 1: Study results

Project name	Sumac cultivation and production project
The proposed project site	**Tafilah governorate**
Project products	**Sumac powder**
Job opportunities	4-6
Total investment (000)	178,879
The net present value of cash flows (000)	134,726
Capital reimbursement period	7
Internal rate of return	%17.02

Estimated capital costs for the project are about (73,829), JD, mostly construction, buildings and storage rooms (18,150) , machinery and equipment (15180), vehicles and transport (16,500) and working capital (21538), where the 50% of the project is supposed to be funded through loans and the rest through self-financing . Despite the project investment cost is not high; it provides employment opportunities that suit the investment magnitude, where it depends on the manual labor in the process of harvesting, processing and marketing.

The result of the financial analysis s indicate that the net present value of the cash flows of the project is positive, which is an indication of the financial feasibility of a project, and the ratio of internal rate of return is much higher than the weighted average cost of capital, which is an indication of the feasibility of the project, other financial indicators show that the project is financially viable and profitable for the investor.

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

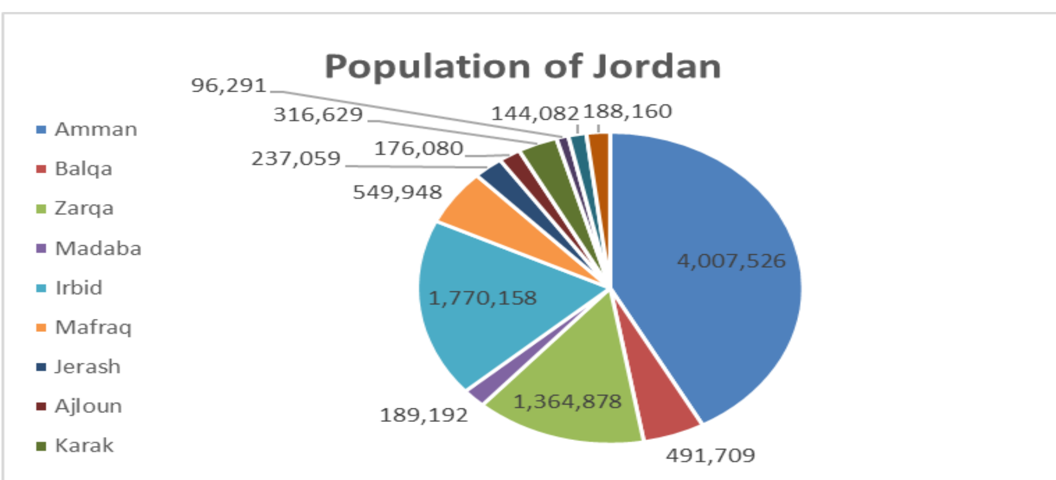
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1:Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 1: Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	**%18.2**
15-64 Years	%61.02	%30.7	**%28.6**
More than 65 Years	%3.94	%1.6	**%1.6**

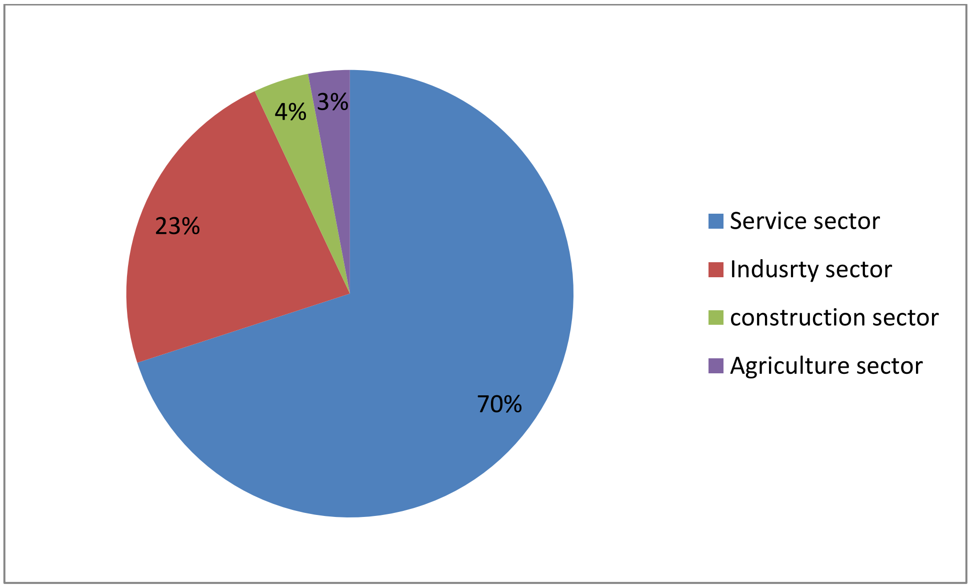
Department of Statistics (DOS), 2015.

Overiew of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2:Sectors Contibution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Invesment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

Investment Climate in Aqaba:

Under the provisions of fifth article of the investment law No. 30 of 2014 and Income tax regulations in the less developed areas, the reduction of the income tax system on the investment projects are clarified in the table below:

Table2 Income Tax and Exemptions

Within the Development Zone or Free Zone	Less Developed Areas
Income Tax is 5%	Income Tax is 20%

	As per Investment Law categorization, Irbid is classified in Class (B) where investors enjoys 80% exemption on Income tax for 20 years, where net income tax after reduction should not be less than 5%.

2. Market analysis

Project Description

Sumac is a wild plant, people didn't seek to plant it previously, because it breeding on its own,it is existing naturally in the mountains of Jerash, Ajloun, Sult, and Tafelieh, and because of the diversity of its food usesthat are: the use of thefruit peel in many local food such as thyme, red and white meat flavors and preparing salads. More recently, demand has increased in commercial markets and prices have risen, prompting some bad people in fraud methods to supply markets.

Sumac is considered one of the plants whose plantation is economically feasible in the local suitable environment in Jordan. There are examples in the neighboring countries on cultivating this plant as the crops in Qalamon in Syria. Sumac trees existence in Jordan are only available on uneven terrain and slopes in enclaves in the Kingdom which enticed many people to think of planting it because the believe that it generates good financial revenues, furthermore it doesn't need an operational cost to serve it or even workers except at harvest time, where farmers used to get rid of it because it grow among trees competing on water and food.

Project Objectives

Due to the increase in sumac prices, and the increase in required quantity year after year, traders heading toward importing sumac from Turkey in order to meet consumers and restaurants needs of sumac that is used in meal preparation as thyme, Al-Msakhan, and salads, therefor; the project aims to produce sumac in Ajloun and market it in the kingdom.

The project aims to:

1- Achieve profit return to the project owners
2- Production and supply of domestic sumac and import it to local market with distinctive quality and brand.
3- Replace local production of sumac rather than importing it.

4- Create jobs in less developed regions, changing the pattern of agricultural production in the area.
5- Encourage farmers to maintain the sumac shrubs that exist in their lands and breeding them as a national crop.

The proposed product

Sumac is used fresh and sprinkled to add flovers to food on regular Salad and particularly (Fattouch),. Moreover, Sumac is considered as a basic material to prepare thyme and the popular dish cooked with chicken and bread (Al-Msakhan).

Sumac is produced from shrubs that turn into bright red in autumn; sumac has a sour flavor where it can be replace Lemon in many recipes. For example in Lebanon and Syria sumac is used to flavor fish, seafood and salads. In Iraq and Turkey sumac is used in chicken, meatballs, kebabs and stews. Also in Iran and Georgia; it can be used to flavor rice and beans.

It is used to give foods sour flavor and good taste, where it is added to thyme to give it the sour flavor.

Sumac is added to chicken, fries, and salads to give them the desired sour flavor. Since sumac contains a lot of cecidozoa material; it is used in leather industry, it was often used in foods but now its use is reduced.

Although it is best to buy full seeds to keep flavor longer, but this is not a practical method, so Sumac seeds must be kept in a well- sealed container away from light and heat, and the project idea is to plant sumac, cultivate it when ripening, fermenting, getting peels, and packaging in plastic bags and containers of different capacities ranging from (100) g to 1 kg

Sumac has three other products in addition to peel:

1- Seeds: it is extracted from the fruit, which is dried completely and crushed.
2- Leaves: taken from branches, which are cut and left to dry completely, then separated from branches, grind and sold.
3- Sour drink (sumac vinegar)
4- The leaves crop: produced from dried leaves that are cut and dried, grind and sold to the dyers and tanners.

Sumac sticks and wood contains yellow pigment and Glycoside that degrades and give (Tannin) which exists also in leaves and flowers to be used in dyeing and leather tanning and medication that prevents bleeding, diarrhea and mucous infections of the mouth and throat.

sumac leaves contain 20-30% tanners substance used in leather tanning , so it becomes the best leather for the manufacture of gloves, binding books and carpet making.

And the other crop is a sour syrup extracted after separation of peel, extracted from each 10 kg fruits about 1 liter sour syrup as sauce or vinegar is added to get vinegar could be called sumac.

Packing:

Plastic bag inside a carton of 80 grams.

Sullivan bag inside small cartons Weight: 250 or 500 grams.

Or 250 g bag without cartons.

Or sumac without packaging

Market size analysis

Department of statistics' data doesn't' indicate any imports description of Sumac as a substance used for food additives, but Jordan consumes large amounts of Sumac and especially sumac seed for food, and there is no statistics showing the quantity or value of sumac produced or consumed by Jordan and not even from theministry of Agriculture, however; imported quantity of sumac was obtained from the General Statistics Department (foreign trade) as it belongs to the category of (other plant products from plant raw materials of a kind used primarily in dyeing and tanning except henna and tattoo) and what is meant from this category is sumac because it is used in tanning and dyeing, so the average import quantity during the period (2011-2015) is about 300 ton/ year worth about 361 000 JD, most of this quantity is imported from Turkey, where 266 tons are imported annually, and since most of it is from Turkey, that indicates that it is for food consumption.

The following table shows the imported and domestic production quantity, and total demand on sumac during the period (2001-2015), where the demand on sumac as was assumed based on the family consumption per year which equals to 250 grams or the average rate per capita around 50 grams, and since the population growth rate in the Kingdom, according to the Department of statistics is about 2.2% per year, it is assumed that the demand growth rate equals to therate of population growth. Domestic demand was expected based on the average per capita consumption estimates during the period (2018-2032) as shown in the following tables:

Table 3: Value and quantity of imports, exports, re-exports and trade deficit of sumac

Year	Value of imports (CIF, 000 JD)	The amount of imports (tons)	Export value (FOB value 000 JD)	National export quantity (tons)	Value of re-exports (FOB JD)	The re-exported quantity (tons)	The total value of exports (JD)	Value of trade deficits (000 JD)
2011	370	296	0	0	36	54	36	-334
2012	336	297	3	3	0	0	3	-333
2013	461	273	1	0	10	8	11	-449
2014	436	222	13	30	0	0	13	-423
2015	204	400	0	0	24	24	24	-180
Average	361	298	3	7	14	17	17	-344

Source: Department of statistics (2015) external trade bulletin.

Since the demand on sumac is increasing, sumac leaves and dried and milled twigs can be sold to increase its income. Since the imported quantity of sumac is about 400 tons in 2015, which means that it is expected that the imported quantities exceed 500 tons in the coming years according to population growth rate of 2.2% plus local production of sumac, which we don't have any statistics for its quantity, where the quality of Jordanian sumac is much better than imported sumac, which is calculated by subtracting net imports and exports from the total demand. So it is expected that sumac will have great demand, based on sumac imported quantities.

Table 4: Evolution of imports and domestic production of sumac (2018 – 2027) ton

Jordan market size for sumac	2011	2012	2013	2014	2015	2016	2017*
Imports (tons)	296	297	273	222	400	408	416
Exports (tons)	0	3	0	30	0	0	0
Re-exports (tons)	54	0	8	0	24	25	26
Domestic production of goods (ton)	100	100	100	100	100	103	105
The demand on commodity (tons)	**342**	**395**	**365**	**292**	**476**	**486**	**496**

Source: Department of statistics (2015), (2) CBE (2015)

Table 5 :Estimate expected demand for sumac (2018 – 2032) ton

The amount of the demand of the commodity	2018	2019	2020	2021	2022	2023	2024	2025
The expected population (000)	10,204	10,428	10,658	10,892	11,132	11,377	11,627	11,883
Annual per capita consumption (kg/capita)	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Projected demand for sumac (tons)	510	521	533	545	557	569	581	594

The amount of the demand of the commodity	2026	2027	2028	2029	2030	2031	2032
The expected population (000)	12,144	12,411	12,684	12,963	13,248	13,540	13,838
Annual per capita consumption	0.05	0.050	0.050	0.050	0.050	0.050	0.050

(kg/capita)							
Projected demand for sumac (tons)	510	621	634	648	662	677	692

Source: study group

Competitors analysis

The local market's needs of sumac are met through import, where large quantities are imported from Turkey, and there are producers on familybusiness level collecting sumac from their lands and market to merchants and spices stores and roaster shops. There are wholesalers specialized in collecting medicinal and aromatic plants where they buy these crops from local producers. There are specialized institutions in packaging spices such as "Alzarqaa' mill" that imports sumac and fill it in small cartons with a capacity of (80) g sold for a fixed price of 0.900 JD in supermarkets.

So the project is not expected to face difficulties in achieving expected sales if it targeted only the local market at competitive prices, so the investor must develop a good marketing plan depends on the preparation and packing sumac free from impurities in clean and attractive packages.

Price analysis and pricing policy

Sumac are sold in the spices shops, supermarkets and hypermarkets either in plastic containers or sealed plastic bags or sold by weight. And the local sumac price reaches in supermarkets and hypermarkets (18-20) JD. Thyme dealers trade sumac for the price of 12 dinars, where Turkish sumac price drops to (6) JD between the producers of thyme.

It is proposed that the price per kilogram of sumac peels is 10 dinars and sumac untreated seeds (4) JD/kg and (1) JD/kg for dry leaves, as a price without transportation where it gives a profit margin to the wholesaler and agro-dealers to be sold for farms with a price ranges between (12-15) JD per kilogram to compete with the importer.

Projected market share

It is suggested to plant 20 donum of sumac in Rocky lands inTafila regions in a rate of one tree for every 5 m^2, approximately 200 sumac shrubs per donum, where the single donum cultivated with sumac produces about 500 kg/year of fruits, which means that the 20 donums farm would produce 10,000 kg/year

As each 1 kg fruits of sumac consists of 40% edible peels, so from every 1 ton of fruits we can get 400 kg of sumac peels, which we get 4.0 ton of sumac peels from 20 donum farm cultivated with sumac, in addition that all of the remaining seed after peeling can be sold as grinded sumac used as food additivesat homes and restaurants, that is we can get 6 tons of seeds powder, that can be converted to smoothies or concentrated syrup, however; each single donum produces about 100 kg of dried leaves equivalent to 2000 kg dried leaves from 20 donums.

Shrub productive rate is 2-3 kg fruits, where one donum of sumac with an age of 3-4 years about 400 kg fruits, if the crust was peeled from the fruit ,then we get 40% of the total fruit weight and sold at a price up to 18 JD/kg .

Expected Revenues Analysis

Sumac production starts on the age of 3 years, where the production of one donum of sumac trees of at the age of 3-4 years about 500 kg increased gradually, and the number of trees is approximately 200 shrubs. The rate of fruit carried on trees is 2.5 kg. Sumac untreated seed are sold at least for about (6) JD/kg.

When the crust is separated from the seed, the collected amount is about half the quantity that is sold in powder form. Production rate per donum is much higher where the price reaches 12 JD/kg, for the purposes of this study, it was considered that the average selling price for sumac crust is (10) JD/kg (4) JD/kg of non-grinded sumac seeds and (1) JD/kg of dry sumac leaves. Sumac retail prices are expected to increase to 2% yearly. Additional revenues from grinded sumac seeds residues were neglected.

Table 6: Sale prices and expected sales total during the period (2018-2033)

Price growth	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Sumac crust	10,000	10,200	10,404	10,612	10,824	11,041	11,262	11,487	11,717	11,951	12,190	12,434	12,682	12,936	13,195	13,459
Sumac cereal and syrup	4,000	4,080	4,162	4,245	4,330	4,416	4,505	4,595	4,687	4,780	4,876	4,973	5,073	5,174	5,278	5,383
Sumac leaves	1,000	1,020	1,040	1,061	1,082	1,104	1,126	1,149	1,172	1,195	1,219	1,243	1,268	1,294	1,319	1,346
Sales of various goods	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**
Sumac crust	0	0	0	21,224	43,297	44,163	45,046	45,947	46,866	47,804	48,760	49,735	50,730	51,744	52,779	53,835
Sumac cereal and syrup	0	0	0	8,490	25,978	26,498	27,028	27,568	28,120	28,682	29,256	29,841	30,438	31,047	31,667	32,301
Sumac leaves				1,061	2,165	2,208	2,252	2,297	2,343	2,390	2,438	2,487	2,536	2,587	2,639	2,692
Total sales	**0**	**0**	**0**	**30,775**	**71,441**	**72,869**	**74,327**	**75,813**	**77,330**	**78,876**	**80,454**	**82,063**	**83,704**	**85,378**	**87,086**	**88,827**

3. Technical analysis

Sumac is considered from the medical plants that have a good scent and strong flavor, despite its weirdness, people like it a lot. Sumac is used as a type of spices, and an alternative of other materials features sour taste. Sumac used in Jordan and Arabic countries in General is from (Rhus coriaria) genus, from (Terebinthaceae) species, and the family of (Anacardiaceae). It is one of more than two hundred and fifty species of the same family which includes cashews, and pistachios.

And usually the length of sumac tree is 3 meters (on average), produce small hairy red berries. This fruit is what Cook usually wants, and it is what sumac is grinded from after seeding and drying.

The importance of sumac and its sour acid flavor and is related to its basic materials such as (Tannins) (organic acids) and oils that exist in large portions in leaves and roots. tannins is considered asan anti-inflammatory used to treat digestive tract ,Ointments and some lotions industry , and more importantly, it plays a big role in dyeing leather. Where tanner's sumac known in Syria contains (Dextrose) and (Mericetine). Moreover, there are other types of sumac known as (Sweet Sumac) scientifically known as (Rhus aromatica) and poison sumac scientifically called (Rhus taxicodendron).

sumac spreads in dry limestone land and moderate tropical environment in general, considered from the wild trees that spreads a lot in Mediterranean region, especially Jordan, Lebanon, Syria, Palestine, the Italian island of Sicily and genrally in southern Italy and Iran. It is from plants and spices which is essential for some oriental cuisines that use it mostly instead of sour substances. It is also often grown in USA ..

1. General description of sumac

Scientific name: Rhus coriaria

Family: (Anacardiaceae)

English name: Sicilian Sumac, Elm-Leaved, and Sumac

2. Home and deployment areas:

Sumac resides in North America and the temperate areas of Asia, the Mediterranean basin and Africa. And naturally exists in Jordan in Sult, Jerash, Ajloun, Tafila, Koura district, Amman, Mahes, and Al-Fheis..

3. The most important types of sumac:

- Tanners' sumac , sumac vinegar (Rhus coriaria)
- (Rhus continues)
- (Rhus tripertata)
- (Rhus aromatic)
- (Rhus taxicodendron)

The first three types are the most important types of sumac in the Mediterranean basin. And newly discovered sumac Rhus tripertata in region of the valley of Shuaib.

4. Uses of sumac:

1- Reduce soil erosion: sumac is considered from the local plants adapted to dry weather conditions and cold climate. It is one of the most important plants that are used to prevent soil erosion, improve their condition and increase the density of the vegetation it is considered from the trees that bear limestone soils.

2- Medical uses: Sumac has many medical uses since antiquity, where ancient Greeks and Arabs used it to treat certain diseases such as gingivitis and treat wounds.

3- Newly, aromatic sumac is used extensively in modern medicine where Gallic acid is effective against bacteria, viruses and fungi.

4- Food uses: fruit peels are used in many local food like thyme, red and white meat flavors and preparing salads.

5- Industrial uses: oils that areextracted from seed can be used in the manufacture of candles. Leaves and water of boiled roots are used as leather tanning.

6- Other uses: used for decorations at public and private parks and on roadsides. Also the flowers of these plants are a source of pollen and nectar for bees.

The used parts of sumac plant:

- Fruit peels – used in food and drug.
- roots peels– used in medical uses
- Leaves – used in tanning.

- Seeding – used to extract oils
- Flowers -- source of pollen for bees

5. Sumac cultivation

Climatic requirements Soil and

1. Temperature: sumac trees bear high temperatures up to 40 C^0, as well as cold and frost, they prefer warm and temperate zones and appropriate temperature of 20-35 C^0 and a minimum temperature of 1-2 C^0, light-loving trees, grow in dry limestone soil or clayey limestone or limestone sand.
2. Rain: Sumac exists in areas that have rainfall ranging from 600 – 1000 mm/year. Sumac is considered from the plants that is draught resistant, where they adapt to rains ranges from 200 – 300 mm/yr. and is durable for snow, wind and dry weather.
3. Height above sea level: Sumac exists at different heights ranging from 0 to 2000 m above sea level. In Jordan the height ranges from 400 to 1400 m above sea level.

It should be taken into consideration when planting sumac it quickly spread in fields and this is a major cause why sumac cultivation is not adopted by most farmers in Jordan where its existence is limited on rough areas and sloping.

6. Methods of propagation of sumac

1- **The seed breeding**: Studies show that sumac proliferation is done through seeding, where one study showed that the highest rate of germination were collected from Anjarah area, where these plants outperformed each of the plants of Ra'as Monef, Ein - Jana, and Eshtafena. As the seed germination studies showed it is better using mechanical methods such as taking off the seed cover, and chemically using concentrated sulphuric acid and algbrilin hormone (BATAINEH, 2006).

Preparing the seed germinating transactions:

- Soak in water for 24 hours at the temperature of 80 – 90 m 2 and leave to cool.
- -Seeds should be soaked in phosphoric acid for 1-6 hours. The duration depends on the type of sumac, as sumac tanners needs a period of not more than 15 minutes where seed germination reaches up to 60%. The best time to seed germination from mid-February to mid-March. Seed can be grown for two months at a temperature of 4 C^0, then seeds are planted

either in bags or nurseries whereit is preferable to use triple mixture of soil, sand and decomposed organic organizer.

- Seedlings are then transported after 6-12 months to permanent land or when seedlings reach to almost half a meter in hight.
- Seedlings are vaccinated to ensure Fruiting since it is a binary housing plant. The vaccination occurs during June or November of the same year, it depends on how strong the seedlings are. To ensure the success of the operation, Species with good fruit characteristics should be multiplied by vegetative and vaccinate seedlings that were produced by seeds.
- There are factors that help the success of seed farming, which is the selection of good seed where seeds are soaked in water, and the deposited seeds are chosen and exclude those that float on water are excluded. Seeds are scratched before planting and then placed at the appropriate depth (3 – 4 cm) with the need to irrigate moderately to avoid rotting.

2- **Vegetative propagation**: Sumac are bred in Autumn after leaves fall off the mother tree aged less than one year old, then transferred to the planting bags or to the permanent land.

3- **Reproducing by transplantation**: it is done after treated with rooting hormone.

4- **Fertilization by irrigation :** Despite plant capacity to withstand drought and frost in many limestone soils, it can be triggered on better growth by fertilization and irrigating at the beginning of planting using super phosphate when planting.

.

- Specifications of the seed planter suitable for transplantation
    - Its thickness to be 7-9 mm.
    - Flexible with high vitality and easily separating bark From wood
    - Intact and doesn't have any pests or diseases, or scratches

Graft specs

- Taken from known, reliable trees.
- Taken from healthy and strong trees.
- Taken at the buds age of 2-8 months.

June is the most appropriate time to conduct transplantation where it is the period where rapid growth occurs for the graft at the highest rate. This period changes depending on climatic conditions. The second date is at Autumn -during Septmber that is known as sleepy eye vaccination.

Success factors for vaccination

- Use clean tools.
- Choosing the right time to be either early morning or before sunset .
- The technician skill responsible for vaccination.

The raw materials needed for production

There is no need for much material and supplies for the production of sumac, most costs are investment costs, but there are costs for fertilizers and supplementary irrigation at the beginning of seedlings plantings to ensure the success of agriculture.

Table 7: Materials and costs required to grow sumac in first year

Production requirements	Required quantity	Price/unit	Total value	Number of donums	Total amount
seedlings	200	0.25	50	20	1,000
Water	300	0.25	75	20	1,500
seedlings planting			20	20	400
Organic fertilizer	2,000	25	50	20	1,000
Total	200	0.50	145	20	3,900

Table 8: Materials and costs for the care of sumac in later years

Compost.	Rate (kg/donum (	price	Value/donum	Number of donums	Total amount
Ammonium sulphate	25	600	15	20	300
Triple super phosphate	60	500	30	20	600
Potassium sulfate	25	500	13	20	250
Magnesium sulfate	5	500	3	20	50

Lesser elements mix	1	2,000	2	20	40
المجموع			62		1,240

Required Human Resources

This project requires an agricultural technician and marketing officer to market sumac and 4 agriculture temporarily workers, and 2 laborers permanently to follow the plant care harvest, preparation of the crop and the treatment operation. The following table shows the required human resources:

Table 9 required human resources

Job title	Number
Direct-staff	
Agricultural production technician	**1**
Marketing Officer	**1**
Total	**2**
Direct employees	
Agriculture workers	2
total	4

الوصف الوظيفي للموظفين :

- **Agricultural Technician**: preparing the land to be suitable for growing crops by tillage, leveling, laying and softening the soil. Provide materials and work equipment of seeds, seedlings, fertilizers and pesticides according to the quantities and specifications required. The development of technical solutions for the problems related to the production process, good knowledge on how to store the product and raw materials, the preparation of technical reports on agricultural operations Production and application of procedures and securing occupational and public safety and health requirements.
- **Marketing and supply representative**: developing the necessary plans for the production, sale and marketing of the products, and receiving the crop and ordering distribution of product by cars to the market places, receipt of raw materials, control records production and sales and revenue and expenses. Contact with buyers, traders and merchants to coordinate sales orders, fill out forms for the receipt and delivery process, follow up the process of loading, unloading of products. negotiating sales prices with customers, responding to customer inquiries and securing their orders, product

sales, customer accounting, invoice editing, warehouse inventory. The mobilization of business models and the application of occupational and public safety and health procedures and instructions.

- **Agricultural worker**: works under the supervision of the agricultural technician and prepares the necessary materials, preparing the land for cultivating by plowing, clearing, preparing the agricultural land, planting the seedlings, caring for the crop, spraying the plants with pest control materials such as insecticides, pesticides and control of diseases widespread and rodents in accordance with environmental standards, , Protecting the crop from the risk of frost. Picking and harvesting products, packaged, sorted and stored in storage houses. Application of occupational safety and health procedures and instructions

Site Analysis

Sumac shrubs breed by seeds, transplantation and suckers so sumac spreads in the land. That's why farmers avoid cultivating sumac in ordinary agricultural lands and prefer planting it in rigged and unproductive lands

Must take into account when selecting the project site the following:

- Provide sufficient quantities of rainfalls
- Preferably choose a neglected territory or rugged and not exploited in in cultivation of fruit trees.
- Tafila is considered a suitable location for this project due to the existence of sufficient affordable lands that provide rugged mountains areas in the governorate.

Based on all these data, it is proposed to establish a facility for sumac production in Tafila in rough and untapped places. Considering , the existence of effective herbal pesticides in destroying sumac shrubs if the farmer wanted to get rid of it when extend to farmlands

Description of production process and manufacturing steps

-Land preparation for planting

- Land is plowed deeply during summer to ensure ventilation

- Organic brewed fertilizers should be added before winter at the rate of 2-5 m^3 per dunam (depends on poverty of the soil) and adding 20-30 kg of phosphate and potash compound fertilizer, can be replaced with organic fertilizers.

- Land is surface plowed to bury chemical fertilizers

- If it is not possible to fertilize the soil because of roughness, direct fertilization by drilling can be done by adding 10 kg organic decomposed fertilizer and about half kg compound fertilizer (phosphate potash) for each hole it should be mixed well with the soil resulting from digging.

- Agriculture

Sumac is cultivated at a distance of 2 m x 2.5 meters between one shrub and the other, seedlings are planted by keeping 15 cm above the soil where the single donum accommodate 200 sumac shrubs.

-Farming operations

Productive trees need to be trimmed to shape the tree structure and remove the tangled weak infected branches to ensure entry of sunlight and wind.

Sumac fruits ripening processes

- The arrival of the fruits to its normal weight and size
- Fruit tend to be Crimson and seed is so hard
- Easiness in separating the crust from the seed.

Harvesting

Manual method:

1. Fruit clusters are gathered and placed in plastic pots where fruits are isolated from clusters, sprinkled with salt and water; for every 10 kg of sumac, a bag and a half of salt are added in 2 liters of water, so sumac is covered with a cloth for 2-3 days without touching it, after this, sumac is uncovered and left to dry.
2. It is placed in a sieve and rubbed to ensure crust separation from seeds where crust remains in the sieve and seeds are thrown away.
3. After harvest is done, it is recommended to leave the crust in the shadow, to maintain the red color and product quality.

Automated method: After picking, sprinkling with salt and water and drying, it is grinded in a special machine, then peels are separated from the seed by a sieve .

Sumac growing stage

The project begins to cultivate sumac in land within a space of to 5 m² per tree, where the distance between one tree and the other (2 m x 2.5 m), either by seed cultivation where to is green in fall and it is a successful method but their growth requires long time compared to seedling cultivation, seeds are grown a week after planting them, or planted in a transplantation method after the treatment with rooting hormone.

Or grow them in seedling method in autumn after separated from the original plant aged less than one year and transferred to land directly, and this is the fastest way, according to experts.

-As the single donum area is 1000 m^2 , then 200 sumac plant can be cultivated per donum.

Production begins in two years and if it is treated as a productive plant, then its productive rate during 8 yearswill be about 500 kg/year

Harvesting and processing of fruits as follows:

- Fruits are picked up as clusters, and seeds are separated from the clusters after exposure to direct solar drying with stirring.
- The seeds are taken to a crust separating machine called Grinder where it is beaded with beaters until crust is separated from the seed.
- Seeds are sieved to separate the crusts which wasn't broken with the machine
- The process should be repeated againto separate the residuer of crust from seed, until 40-50% of the weight of the seeds are separated prior to the start of process.
- Peels are taken, grinded and packaged for sale or sold as traders' requests
- Sour syrup is extracted from the remaining seed after separation, then it is concentrated in dark bottles with the capacity of 0.3 liters per bottle
- Dried leaves are sold to tanners and the dyers

Technical Requirements Analysis

Technical requirements can be summarized for the project using drip irrigation systems at the beginning of the project only as follows:

- Area of land within 20 donums
- Installing drip irrigation systems with full equipment and pumps.
- Digging pits and adding organic fertilizers t
- Planting sumac seedlings after bred in nursery
- Tree care for three years.
- Collect, ferment and dry the crop. Store in storage
- Grinding the seeds to get the sumac crust
- Product packing in vacuumed bags

Land and Construction Works

project buildings will be established on a 300 donum 100 square meters devoted toadministration offices and stores, and the remaining space devoted to services, facilities and trails and it was assumed that the farm land area of 20 donum for 20 JD/ donum,and fees increases at the rate of 2% which is included in the operating costs. The following table shows the required plot

Table10 : required land

Land	(Cost) JD/m 2	Area (m2)
[1] **Required land**	20	300

Construction work required for the project is as follows

Table 11: required construction work

Construction work	Area m2
Administrative offices	**80**
Stores	**40**
Production hall	**150**
Land preparation	
Other works	

Machinery and equipment

Most of the machines, and necessary equipment's are available in the local market, the following table shows the machinery and equipment needed for the project:

Table 12: Machines and equipment

Machinery and equipment	Quantity
Agricultural plants	4000
hammers mill	1
Cylindrical Sieve with paddles.	1
stainless tank for soaking with a strainer in bottom 500-liter	1

capacity engine	
Packaging machine	1
Scales (balances Pecan filling)	1
Generator	1
Deep plows	3
Agricultural tools and equipment	1
Drip irrigation system	20

Furniture and furnishings

The project does not need Office furniture but some simple furniture such as a table and 4 plastic chairs, and a table during the harvest, so the project needs the following furniture and furnishings

Table 13: required furniture and furnishings

Furniture and furnishings	Quantity
Office furniture	1

Vehicles and transportation

The following table shows the vehicles and transportation meansrequired for the project:

Table 14: Vehicles and transportation

Vehicles and transportation	Quantity
Pick-up	**1**

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

Sumac cultivation and production - Tafilah

- The weighted average cost of capital used in the study is 11.97% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Icome tax on the net project income is 0 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 15 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumption

Table 15 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2033

Table 16 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 17 working time for the project

Working time	
No. of weeks	52
No. of days per week	6
No. of days annually	312
No. of working hours	8

Table 18 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	2.00%
Annual Expenses Increase Rate	2.00%
Annual Increase Rate in Employees Salaries	3.0%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 19 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	0.00%
Utilities Expenses from Total Revenues	0.00%
Maintenance Expenses from Total Revenues	1.00%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.00%
Marketing Expenses from Total Revenues	2.00%

Table 20 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	5%
(2) Income Tax Deduction	100%
Income Tax after Deduction	0%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	**Value**
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	14

Table 21 Risk Premuim

Risk Premuim	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	7.50%
(3) Market Risk Premuim	9.93%

Income Tax Rate	0.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 22 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%7.50
Market Risk Premuim	%9.93
Income Tax Rate	%0
Bets	1.00
Equity	89,439
Loans Value	89,439
Loans	
Cost of Borrowing before Tax	%7.50
Cost of Borrowing After Tax	%7.50
Loans Value	89,439
Loans Percentage	%50
Equity	
Cost of Equity	%16.4
Equity Value	89,439
Equety Percentage	%50
Gross	
Project Value	178,879
Weighted Average Cost of Capital	**%11.97**

Capital Expenditures

The estimated cost of the project is 162,617 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 23Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost(JD)
Required Land	20	300	6,000
Total Cost			**6,000**

Table 24 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross / Cost
Administrative offices	100	80	8,000
Stores	150	40	6,000
Production hall	150	150	22,500
Land preparation		150	1,000
Other works	-	-	3,750
Total			**41,250**

Table 25 Machinery and equipment Capital Expenditure

Machinery and equipment	Number	Unit cost (JD)	Total cost (JD)
Agricultural plants	4000	0.250	1000
hammers mill	1	3,000	3000
Cylindrical Sieve with paddles.	1	2,500	2500
stainless tank for soaking with a strainer in bottom 500-liter capacity engine	1	1,500	1500
Packaging machine	1	500	500
Scales (balances Pecan filling)	1	400	400
Generator	1	500	500

Deep plows	3	300	900
Agricultural tools and equipment	1	500	500
Drip irrigation system	20	150	3,000
Total cost			**13,800**

Table 26 Furniture Capital Expenditure

Furniture	Number	Unit cost (JD)	Total cost (JD)
General furniture	1	1,000	1,000
Total			**1,000**

Table 27 Transporation and Vehicles Capex

Transporation and Vehicles	No	Unit Cost/JOD	Total Cost/JOD
Medium vehicle	1	15,000	15,000
Total cost	**1**		**15,000**

Table 28 Pre-Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	0
Transportation	0
Marketing	500
Training	500
Total cost	**1,000**

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 29 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	36	%300	57,882
General and administrative expenses	36	%300	1,485

Marketing expenses	36	%300	0
Indirect Salaries	36	%300	25,200
Total			**84,567**

Table 30 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	6,000	10%	6,600	%3.7
Construction works	41,250	10%	45,375	%25.4
Machinery &Equipment	13,800	10%	15,180	%8.5
Furniture	1,000	10%	1,100	%0.6
Vehicles	15,000	10%	16,500	%9.2
Pre-Operating Expenses	1,000	10%	1,100	%0.6
Working capital	84,567	10%	93,024	%52.0
Total (JD)	**162,617**		**178,879**	**100%**

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (178,879) JD, where 50% of the project will be financed through loans (89,439) JD, and the remaining (50%) through (Equity) with (89,439) JD. The following table summarizes the general structure of the required financing:

Table 31 Sources of Funding

Funding Sources	Amount	Percentage
Equity	89,439	%50
Loans	89,439	%50
Total	**178,879**	**%100**
Use of Fund	**Amount**	**Percentage**
Capital Expenditures	84,755	%47.38
Pre-Operating Expenses	1,100	%0.61
Working Capital	93,024	%52.00
Total	**178,879**	**%100.00**

Operating Expenses

The following table sumarizes the expected operating expenses for the proposed project (2018-2033):

Table 32 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Raw Materials and water from Revenues	5,140	1,240	1,240	1,240	1,240	1,240	1,240	1,240	1,240	1,240	1,240	1,240	1,240	1,240	1,240	1,240
Agricultural land renting	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400	400
Maintenance from Revenues	0	0	389	768	784	799	815	832	848	865	883	900	918	937	955	974
Disposables from Revenues	0	0	389	768	784	799	815	832	848	865	883	900	918	937	955	974
Direct Salaries	12,000	7,210	7,426	7,649	7,879	8,115	8,358	8,609	8,867	9,133	9,407	9,690	9,980	10,280	10,588	10,906
Others	1,754	885	984	1,083	1,109	1,135	1,163	1,191	1,220	1,250	1,281	1,313	1,346	1,379	1,414	1,449
Total	19,294	**9,735**	**10,829**	**11,908**	**12,195**	**12,489**	**12,792**	**13,104**	**13,424**	**13,754**	**14,094**	**14,443**	**14,802**	**15,172**	**15,553**	**15,944**

General and Administrative Expenses

The following table sumarizes the expected general and administrative expenses for the proposed project (2018-2033):

Table 33 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Telecom	50	51	52	53	54	55	56	57	59	60	61	62	63	65	66	67
Consulting	400	408	416	424	433	442	450	459	469	478	488	497	507	517	528	538
Miscellaneous	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	61
Total	495	**505**	**515**	**525**	**536**	**547**	**557**	**569**	**580**	**592**	**603**	**615**	**628**	**640**	**653**	**666**

Marketing Expenses

Table 34 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Marketing Expenses	**386**	**195**	**217**	**238**	**244**	**250**	**256**	**262**	**268**	**275**	**282**	**289**	**296**	**303**	**311**	**319**

Human Resources

The total annual salaries and monthly wages has been estimated on 14 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 35 expected number of employees

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Indirect employees																
Agricultural production	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Marketing Officer	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1
Total number of indirect	**1**	**1**	**1**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**
direct employees																
Agriculture workrs	4	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Total number	**4**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**

Table 36 Expected monthly Salaries and Wages (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Indirect employees																
Agricultural production	600	618	637	656	675	696	716	738	760	783	806	831	855	881	908	935
Marketing Officer	400	412	424	437	450	464	478	492	507	522	538	554	570	587	605	623
direct employees																
Agriculture workrs	250	258	265	273	281	290	299	307	317	326	336	346	356	367	378	389

Table 37 Expected Annual Salaries and Wages (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Indirect employees																
Agricultural production	8,400	8,652	8,912	9,179	9,454	9,738	10,03	10,33	10,64	10,96	11,28	11,62	11,97	12,33	12,70	13,08
Marketing Officer	0	0	0	6,119	6,303	6,492	6,687	6,887	7,094	7,307	7,526	7,752	7,984	8,224	8,471	8,725
Total indirect salaries	**8,400**	**8,652**	**8,912**	**15,29**	**15,75**	**16,23**	**16,71**	**17,21**	**17,73**	**18,26**	**18,81**	**19,37**	**19,96**	**20,55**	**21,17**	**21,81**
direct employees																
Agriculture workrs	12,000	7,210	7,426	7,649	7,879	8,115	8,358	8,609	8,867	9,133	9,407	9,690	9,980	10,280	10,588	10,906
Total direct salaries	12,000	**7,210**	**7,426**	**7,649**	**7,879**	**8,115**	**8,358**	**8,609**	**8,867**	**9,133**	**9,407**	**9,690**	**9,980**	**10,280**	**10,588**	**10,906**
Total salaries	20,400	**15,862**	**16,338**	**22,947**	**23,636**	**24,345**	**25,075**	**25,827**	**26,602**	**27,400**	**28,222**	**29,069**	**29,941**	**30,839**	**31,764**	**32,717**

Depreciations

The following tables illustrate cost of Capex and Depreciaition rates:

Table 38 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	6,600	%0.0	%0.0
Construction Works	45,375	%0.1	%0.0
Machinaries	15,180	%7.0	%3.0
Furniture	1,100	%7.0	%4.0
Vehicles	16,500	%7.0	%5.0
	84,755		

Table 39 Depreciations and Additions on Construction Works

Capex, Annual Additions and depreciaitions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Total Fixed Assets	84,755	86,079	87,460	88,901	90,403	91,970	93,604	95,309	97,088	98,944	100,881	102,902	105,011	107,212	109,510	111,908
Total Depreciation Expenses	2,276	2,367	2,462	2,562	2,665	2,773	2,886	3,003	3,126	3,254	3,388	3,527	3,673	3,824	3,983	4,149
Total Accumulated Depreciation	2,276	4,643	7,106	9,667	12,332	15,105	17,991	20,994	24,120	27,374	30,762	34,289	37,961	41,786	45,769	49,918
Total Additions	0	1,324	1,381	1,440	1,502	1,567	1,634	1,705	1,779	1,856	1,937	2,021	2,109	2,201	2,298	2,398
Total Net Book Values	82,479	81,436	80,355	79,234	78,071	76,864	75,613	74,315	72,968	71,570	70,119	68,613	67,049	65,426	63,741	61,991

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 40 Loan Details

Loan Value	89,439
Annual Interst Rate	9.50%
Loans Period	8
Grace Period	0
Loan Starts at year:	2018

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				89,439
2019	16,461	8,497	7,964	81,475
2020	16,461	7,740	8,721	72,754
2021	16,461	6,912	9,549	63,205
2022	16,461	6,004	10,456	52,749
2023	16,461	5,011	11,450	41,299
2024	16,461	3,923	12,538	28,761
2025	16,461	2,732	13,729	15,033
2026	16,461	1,428	15,033	0

Income Statement

Table 41 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Revenues																
Sales Revenues	0	0	38,911	76,831	78,368	79,935	81,534	83,165	84,828	86,525	88,255	90,020	91,821	93,657	95,530	97,441
Gross Operating	0	0	38,911	76,831	78,368	79,935	81,534	83,165	84,828	86,525	88,255	90,020	91,821	93,657	95,530	97,441
Operating Expenses	19,294)	(9,735)	10,829)	11,908)	12,195)	12,489)	12,792)	13,104)	13,424)	13,754)	14,094)	14,443)	14,802)	15,172)	15,553)	15,944)
Gross Operating	19,294)	(9,735)	28,082	64,923	66,174	67,446	68,742	70,061	71,404	72,770	74,161	75,577	77,018	78,485	79,978	81,497
Gross Profit	*DIV/0#*	*DIV/0#*	*%72*	*%85*	*%84*	*%84*	*%84*	*%84*	*%84*	*%84*	*%84*	*%84*	*%84*	*%84*	*%84*	*%84*
Salaries and Benefits	(8,400)	(8,652)	(8,912)	15,298)	15,757)	16,230)	16,717)	17,218)	17,735)	18,267)	18,815)	19,379)	19,961)	20,559)	21,176)	21,812)
General and	(495)	(505)	(515)	(525)	(536)	(547)	(557)	(569)	(580)	(592)	(603)	(615)	(628)	(640)	(653)	(666)
Markeing Expenses	0	0	(217)	(238)	(244)	(250)	(256)	(262)	(268)	(275)	(282)	(289)	(296)	(303)	(311)	(319)
Pre Operating	(1,100)															
Gross Indirect	(9,995)	(9,157)	(9,643)	16,062)	16,537)	17,026)	17,530)	18,049)	18,583)	19,133)	19,700)	20,284)	20,884)	21,503)	22,140)	22,797)
Income before	29,289)	18,892)	18,439	48,862	49,637	50,420	51,212	52,012	52,821	53,637	54,461	55,294	56,134	56,982	57,837	58,700
Fixed Assets	(2,276)	(2,367)	(2,462)	(2,562)	(2,665)	(2,773)	(2,886)	(3,003)	(3,126)	(3,254)	(3,388)	(3,527)	(3,673)	(3,824)	(3,983)	(4,149)
Income before Tax	31,565)	21,259)	15,977	46,300	46,972	47,647	48,326	49,009	49,695	50,383	51,074	51,767	52,461	53,157	53,854	54,552
Bank Interests	(8,497)	(7,740)	(6,912)	(6,004)	(5,011)	(3,923)	(2,732)	(1,428)	0	0	0	0	0	0	0	0
Income Before Tax	40,062)	28,999)	9,065	40,295	41,961	43,724	45,594	47,581	49,695	50,383	51,074	51,767	52,461	53,157	53,854	54,552
Income Tax	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Net Profit	40,062)	28,999)	9,065	40,295	41,961	43,724	45,594	47,581	49,695	50,383	51,074	51,767	52,461	53,157	53,854	54,552
Net Profit Percentage	*DIV/0#*	*DIV/0#*	*%23*	*%52*	*%54*	*%55*	*%56*	*%57*	*%59*	*%58*	*%58*	*%58*	*%57*	*%57*	*%56*	*%56*
Compulsory Reserves	0	0	(906)	(4,030)	(4,196)	(4,372)	(4,559)	(4,758)	(4,969)	(5,038)	(5,107)	(5,177)	(5,246)	(5,316)	(5,385)	(5,455)
Retained Earnings	40,062)	69,061)	60,903)	24,637)	13,128	52,479	93,514	136,33	181,06	226,40	272,37	318,96	366,17	414,02	462,48	511,58

Expected Cashflows Statement

The table belwo summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 42 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Cash Inflows from Operating Activities																
Net Profit	(40,062)	(28,999)	9,065	40,295	41,961	43,724	45,594	47,581	49,695	50,383	51,074	51,767	52,461	53,157	53,854	54,552
Bank Interests	8,497	7,740	6,912	6,004	5,011	3,923	2,732	1,428	(0)	(0)	0	0	0	0	0	0
Depreciation	2,276	2,367	2,462	2,562	2,665	2,773	2,886	3,003	3,126	3,254	3,388	3,527	3,673	3,824	3,983	4,149
Total Operating Cashflows before Additions to Working Capital	(29,289)	(18,892)	18,439	48,862	49,637	50,420	51,212	52,012	52,821	53,637	54,461	55,294	56,134	56,982	57,837	58,700

Inventory (Increase/Decrease)	(857)	650	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Accounts Receivables (Increase/Decrease)	0	0	(3,243)	(3,160)	(128)	(131)	(133)	(136)	(139)	(141)	(144)	(147)	(150)	(153)	(156)	(159)
Accounts Paybales (Increase/Decrease)	1,180	(472)	91	90	24	25	25	26	27	27	28	29	30	31	32	33
Working Capital (Increase/Decrease)	323	178	(3,151)	(3,070)	(104)	(106)	(108)	(110)	(112)	(114)	(116)	(118)	(120)	(122)	(124)	(127)
Net Cashflows from Operating Activities	(28,966)	(18,713)	15,287	45,791	49,533	50,314	51,104	51,902	52,709	53,523	54,345	55,176	56,014	56,859	57,713	58,574
Cashflows from Investments Activities																
Fixed Assets (Procurement)	(84,755)	(1,324)	(1,381)	(1,440)	(1,502)	(1,567)	(1,634)	(1,705)	(1,779)	(1,856)	(1,937)	(2,021)	(2,109)	(2,201)	(2,298)	(2,398)
Net Cashflows from Investments Activities	(84,755)	(1,324)	(1,381)	(1,440)	(1,502)	(1,567)	(1,634)	(1,705)	(1,779)	(1,856)	(1,937)	(2,021)	(2,109)	(2,201)	(2,298)	(2,398)
Cashflows from Financing Activities																

Capital	89,439															
Loan Amortization	(7,964)	(8,721)	(9,549)	(10,456)	(11,450)	(12,538)	(13,729)	(15,033)	(0)	(0)	0	0	0	0	0	0
Bank Interest Rate	(8,497)	(7,740)	(6,912)	(6,004)	(5,011)	(3,923)	(2,732)	(1,428)	0	0	0	0	0	0	0	0
Loans	89,439	0	0	0	0	0	0	0	0	0	1	2	3	4	5	6
Net Cashflows from Financing Activities	162,418	(16,461)	(16,461)	(16,461)	(16,461)	(16,461)	(16,461)	(16,461)	0	0	1	2	3	4	5	6
Net (Increase/Decrease) in Cash	48,697	(36,499)	(2,555)	27,890	31,569	32,287	33,009	33,736	50,930	51,667	52,410	53,157	53,908	54,662	55,420	56,181
Cashflows at the Beginning of Period	0	48,697	12,198	9,643	37,533	69,103	101,389	134,398	168,135	219,065	270,732	323,141	376,298	430,206	484,868	540,288
Cashflows at the End of Period	48,697	12,198	9,643	37,533	69,103	101,389	134,398	168,135	219,065	270,732	323,141	376,298	430,206	484,868	540,288	596,469

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 43 Expected balance sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Assets																
Current Assets																
Cash	48,697	12,198	9,643	37,533	69,103	101,389	134,398	168,135	219,065	270,732	323,141	376,298	430,206	484,868	540,288	596,469
Inventory	857	207	207	207	207	207	207	207	207	207	207	207	207	207	207	207
Accounts Receivable	0	0	3,243	6,403	6,531	6,661	6,795	6,930	7,069	7,210	7,355	7,502	7,652	7,805	7,961	8,120
Total current Assets	49,553	**12,404**	**13,093**	**44,143**	**75,840**	**108,257**	**141,400**	**175,272**	**226,340**	**278,149**	**330,703**	**384,006**	**438,064**	**492,879**	**548,455**	**604,796**
Non Current Assets																
Fixed Assets (net)	82,479	81,436	80,355	79,234	78,071	76,864	75,613	74,315	72,968	71,570	70,119	68,613	67,049	65,426	63,741	61,991
Total Non Current Assets	82,479	**81,436**	**80,355**	**79,234**	**78,071**	**76,864**	**75,613**	**74,315**	**72,968**	**71,570**	**70,119**	**68,613**	**67,049**	**65,426**	**63,741**	**61,991**
Total Assets	132,032	**93,841**	**93,447**	**123,376**	**153,911**	**185,122**	**217,013**	**249,587**	**299,308**	**349,718**	**400,821**	**452,619**	**505,113**	**558,305**	**612,196**	**666,786**
Liabilities																
Current Liabilities																
Paybles	1,180	708	799	889	913	937	963	989	1,015	1,043	1,071	1,100	1,130	1,161	1,193	1,225
Remaining amount of Loan	8,721	9,549	10,456	11,450	12,538	13,729	15,033	0	0	0	0	0	0	0	0	0
Total current Liabilities	9,900	**10,257**	**11,256**	**12,339**	**13,450**	**14,666**	**15,995**	**989**	**1,015**	**1,043**	**1,071**	**1,100**	**1,130**	**1,161**	**1,193**	**1,225**
Non Current Liabilities																
Long Terms Loans	72,754	63,205	52,749	41,299	28,761	15,033	(0)	(0)	(0)	(0)	0	0	0	0	0	0
Total Long Term	72,754	**63,205**	**52,749**	**41,299**	**28,761**	**15,033**	**(0)**	**(0)**	**(0)**	**(0)**	**0**	**0**	**0**	**0**	**0**	**0**

Liabilities																
Total Liabilities	82,655	**73,462**	**64,004**	**53,638**	**42,212**	**29,699**	**15,995**	**989**	**1,015**	**1,043**	**1,071**	**1,100**	**1,130**	**1,161**	**1,193**	**1,225**
Owners Equity																
Shareholders Contributions	89,439	89,439	89,439	89,439	89,439	89,439	89,439	89,439	89,439	89,439	89,439	89,439	89,439	89,439	89,439	89,439
Statutory Reseve	0	0	906	4,936	9,132	13,504	18,064	22,822	27,791	32,830	37,937	43,114	48,360	53,676	59,061	64,516
Retained Profits	(40,062)	(69,061)	(60,903)	(24,637)	13,128	52,479	93,514	136,337	181,062	226,406	272,373	318,963	366,178	414,020	462,488	511,585
Total Equity	49,378	20,378	29,443	69,739	111,699	155,423	201,017	248,598	298,293	348,676	399,749	451,516	503,977	557,134	610,989	665,540
Total Liabilities and Equity	132,032	**93,841**	**93,447**	**123,376**	**153,911**	**185,122**	**217,013**	**249,587**	**299,308**	**349,718**	**400,820**	**452,616**	**505,107**	**558,295**	**612,181**	**666,765**

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 44 Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	Final value
Net Free Cashflows	(178,879)	(27,866)	(20,038)	13,906	44,351	48,030	48,747	49,470	50,197	50,930	51,667	52,409	53,155	53,905	54,658	55,415	674,623	(178,879)

Discount Factor	1.00	0.89	0.80	0.71	0.64	0.57	0.51	0.45	0.40	0.36	0.32	0.29	0.26	0.23	0.21	0.18	0.18	1.00
Net Present Value for Cash Flows	(178,879)	(24,888)	(15,984)	9,908	28,221	27,296	24,743	22,427	20,325	18,418	16,687	15,118	13,695	12,404	11,233	10,172	123,830	(178,879)

The table below illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 45 Payback Period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	Final value
Net Free Cashflows	(178,879)	(27,866)	(20,038)	13,906	44,351	48,030	48,747	49,470	50,197	50,930	51,667	52,409	53,155	53,905	54,658	55,415	674,623	(178,879)
Project Value Returned	178,879	206,745	226,783	212,876	168,525	120,495	71,747	22,278	(27,920)	(78,850)	(130,517)	(182,925)	(236,080)	(289,985)	(344,643)	(400,058)	(1,074,681)	178,879
Payback Period (Year)	7	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	7

Table 46 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%11.97
Net Present Value for Cashflows	134,726
Payback Period	7
Internal Rate of Return	%17.02

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 47 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC	Sensitivity Analysis
Original Scenario	18.58%	7	120	11.97%
Revenues declined by 10%	16.09%	8	72	11.97%
Operating Expenses Increased by 10%	17.66%	8	105	11.97%
Total cost Increased by 10%	17.03%	8	97	11.97%
Total cost Increased and revenues decreased by 10%	14.53%	9	49	11.97%

Conclusions and Recommendations

Based on the profitability analysis and considering that project inrernal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 134,726 Jordan Dinars considering that the project provides 4-6 Job Opportunitites for the governorate residents.